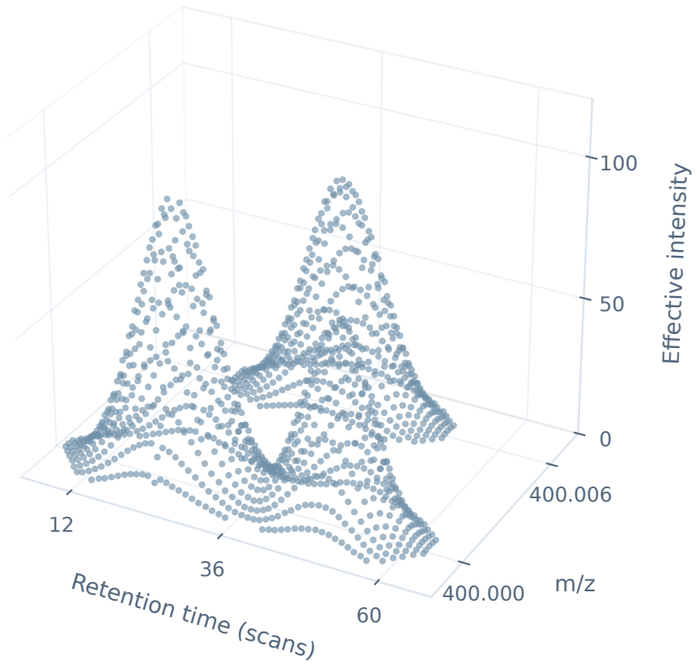
A basin is a cloud of points that leads to one apex

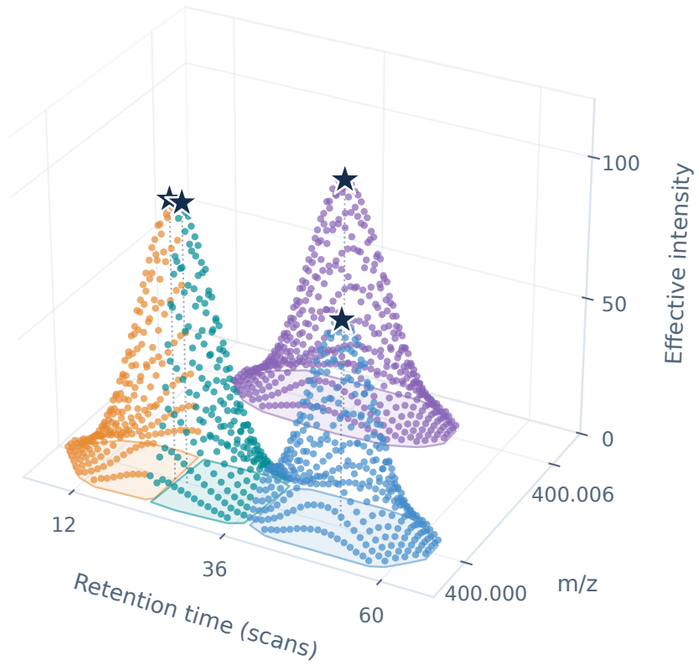
Each dot is a centroid. Colour shows basin membership; a star marks its apex.

1 Start with a point cloud



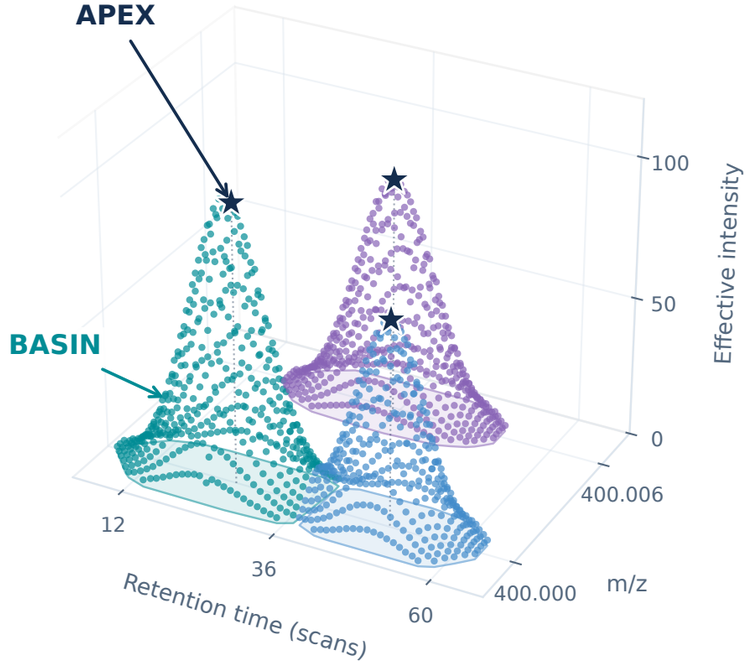
Positions: retention time and m/z.
Height: effective intensity.

2 Assign points to apexes



Four initial basins.
The shallow dip creates an extra apex.

3 Merge the shallow split



Three surviving basins.
The weak split joins its neighbour.

Same colour = one basin • ★ = apex • Very low-intensity points omitted for clarity

BASIN = points sharing an apex. **APEX = the endpoint of their uphill paths.**

Illustrative synthetic clouds with broadened m/z spread • memberships computed by Peak3D • faint patches = projected footprints